



[Course](#) > [4. Reporting your findings](#) > [4.6 Writing reports using LaTeX](#) > Mathematical formulas in LaTeX

Mathematical formulas in LaTeX

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LaTeX has the capability to typeset mathematical formulas and equations without having to use an equation editor. Instead, **LaTeX** makes use of commands and symbols to typeset the formulas.

On this page we will introduce you to the most common building blocks of mathematics in **LaTeX**. You will practice with this in the exercises. In the next page, we will discuss how you can include these formulas in your report.

Powers, exponents, superscripts and subscripts

Mathematics contain often powers of numbers, exponentials, superscripts and subscripts. Examples of these are e^t , x^2 and X_1 .

In **LaTeX**, you can obtain this by using the `^` and the `_` symbol. The three examples above are generated as follows:

- `e^t` results in e^t .
- `x^2` results in x^2 .
- `X_1` results in X_1 .

It might happen you want more than one symbol or letter to display as a superscript or subscript. In this case you need to group all symbols together using curly braces, like in the next examples:

- `e^{-0.7t}` results in $e^{-0.7t}$.
- `X_{1,\text{eq}}` results in $X_{1,\text{eq}}$.

Note that letters are displayed slanted by **LaTeX** by default in mathematics. If you want to include non-slanted letters, such as **eq** in the example, you can use the command `\text{...}`.

Different uses of formulas

The code for the mathematics above is not directly usable: in **LaTeX** mathematical formulas must be written in *math mode*. **LaTeX** recognises you want math mode by using some special commands before and after the formula.

Within a scientific report, you can think of several places and ways to display mathematics. Three of these are:

1. In the text as part of a sentence;
2. As an unnumbered equation;
3. As a numbered equation.

We will discuss each of this in the following.

Inline formulas

Mathematics in the text as part of a sentence is called *inline math* in **LaTeX**. To add such mathematics, you must put your mathematical code between special symbols. There are two options for this:

- Put the code between `$` and `$`;
- Put the code between `\(` and `\)`.

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An example of an inline formula is $\frac{dP}{dt} = 0.7 P(t)$.

is compiled as

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by **LaTeX**. As you can see, the fraction is displayed relatively small. If you want bigger symbols, you can add `\displaystyle` after the opening symbol of the inline math. This results in

An example of an inline formula is $\frac{dP}{dt} = 0.7 P(t)$.

Unnumbered equations

If you want to draw more attention to a certain formula, you can give the formula its own line, like this:

$$\frac{dP}{dt} = 0.7P(t) - 0.007P^2(t) - 0.04P(t)G(t).$$

To achieve this in **LaTeX**, again, three options exist:

- Put the code between `\[` and `\]`;
- Put the code between `$$` and `$$`;
- Put the code between `\begin{equation*}` and `\end{equation*}`;

Unnumbered equations are automatically in *display style*, so if you want smaller fractions, you can use the command `\tfrac` instead of `\frac` or `\dfrac` (see also below).

Numbered equations

If you want to draw more attention to a certain formula, and also want to refer back to it (it is that important), you should put a number behind the equation. **LaTeX** will do this automatically if you use the following option:

- Put the code between `\begin{equation}\label{mytag}` and `\end{equation}`.

If it is the second numbered equation in Chapter 3, this results in:

$$\frac{dP}{dt} = 0.7P(t) - 0.007P^2(t) - 0.04P(t)G(t).$$

If you want to refer back to this equation in your text, use the command `\ref` in your text. So `\ref{mytag}` gives you **3.2** in the text. If you want to have parentheses around the number, you can use `\eqref` instead.

Numbered equations are, just like unnumbered equations in display style.

Fractions, roots and vectors

LaTeX can also typeset fractions, roots and vectors:

- `\frac{3}{4}` results in $\frac{3}{4}$.
- `\sqrt{a^2+b^2}` results in $\sqrt{a^2 + b^2}$.
- `\vec{X}` results in $\vec{X}$.

As you can see, you can combine all commands to produce intricate and complex formulas.

Derivatives

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- `\frac{d\vec{x}}{dt}` results in $\frac{d\vec{x}}{dt}$ in text and in (un)numbered equations as:

$$\frac{d\vec{x}}{dt}.$$

- `\frac{\partial f_1}{\partial G}` results in $\frac{\partial f_1}{\partial G}$.

If you want to have a larger fraction in your text, you can use the `\dfrac{...}{...}` command:

- `\dfrac{d\vec{x}}{dt}` results in $\frac{d\vec{x}}{dt}$.
- `\dfrac{\partial f_1}{\partial G}` results in $\frac{\partial f_1}{\partial G}$.

If you want to have a smaller fraction in a(n) (un)numbered equation, you can use the `\tfrac{...}{...}` command:

$$\tfrac{d\vec{x}}{dt}.$$

Integrals

Integrals can be created in **L^AT_EX** with the use of the command `\int`:

- In the text, the command `\int_a^b f(x)dx` results in $\int_a^b f(x)dx$.
- In (un)numbered equations, this gives:

$$\int_a^b f(x)dx.$$

- The limits of the integral can also be placed above and below the integral instead of to the right, by using `\int\limits_a^b f(x)dx`:

$$\int\limits_a^b f(x)dx.$$

Exercise A

1/1 point (graded)

Now you know a little about mathematics in **L^AT_EX**, try to find the command to create the following unnumbered equation:

$$\frac{dP}{dt} = 0.007P^2(t) - 0.04P(t)G(t) + \sqrt{G(t) - \int_0^T P(t)dt}.$$

The command to generate the formula is:

Remove all unnecessary spaces and curly brackets from your answer, and put all commands on one line.

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$$\frac{dP}{dt} = 0.007P^2(t) - 0.04P(t)G(t) + \sqrt{G(t) - \int_0^T P(t)dt}.$$



Answer: $\frac{dP}{dt} = 0.007P^2(t) - 0.04P(t)G(t) + \sqrt{G(t) - \int_0^T P(t)dt}.$ or $\frac{dP}{dt} = 0.007P^2(t) - 0.04P(t)G(t) + \sqrt{G(t) - \int_0^T P(t)dt}.$ or

$$\frac{dP}{dt} = 0.007P^2(t) - 0.04P(t)G(t) + \sqrt{G(t) - \int_0^T P(t)dt}.$$

or

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or

$$\frac{dP}{dt} = 0.007P^2(t) - 0.04P(t)G(t) + \sqrt{G(t) - \int_0^T P(t)dt}.$$

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You have used 15 of 30 attempts

i Answers are displayed within the problem

Special functions and brackets

If you use special functions such as the sine function or a limit in your formula, you must typeset these upright. **L^AT_EX** helps you with this by providing several extra commands:

- $\sin(\frac{3}{4})$ results in **$\sin(\frac{3}{4})$** .
- $\lim_{x \rightarrow 0} \cos(x)$ results in **$\lim_{x \rightarrow 0} \cos(x)$** .

The brackets in the first example are a little too small and you might prefer the "x → 0" to be positioned under the symbol for the limit. You can force this by the commands $\left(, \right)$ and $\limits$:

- $\sin\left(\frac{3}{4}\right)$ results in **$\sin\left(\frac{3}{4}\right)$** .
- $\lim\limits_{x \rightarrow 0} \cos(x)$ results in **$\lim\limits_{x \rightarrow 0} \cos(x)$** .

If you want another bracket than (or), you can replace it with the command for that bracket. The command $\left$ must always be accompanied by the command $\right$, so if you want to have only one big bracket, you can use the command $\left$ or $\right$ to create an empty bracket for the side you want to skip:

- $\left\{\frac{3}{4}\right\}$ gives **$\left\{\frac{3}{4}\right\}$** .
- $\left.\frac{\partial f_1}{\partial P}\right|_{(P_0, G_0)}$ gives **$\left.\frac{\partial f_1}{\partial P}\right|_{(P_0, G_0)}$** .

Systems and matrices

In this course you have seen the system of differential equations

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$$\begin{cases} \frac{dP}{dt} = 0.7P(t) - 0.007P^2(t) - 0.04P(t)G(t), \\ \frac{dG}{dt} = -0.25G(t) + 0.008P(t)G(t). \end{cases}$$

The alignment of the left hand side, equal sign and the right hand sides can be achieved by using arrays in **L^AT_EX**. The commands to create such a system are:

```
\left{ % The left brace
\begin{array}{rcl} % An array with 4 columns, (right aligned, centered, left aligned).
% First row. Elements in a row are separated by &. \dfrac is \frac in displaystyle
\dfrac{dP}{dt} & = & 0.7 P(t) - 0.007 P^2(t) - 0.04P(t)G(t), \\ % new row in array
\\ % An extra row for some spacing
\dfrac{dG}{dt} & = & -0.25G(t) + 0.008 P(t)G(t).
\end{array}
\right.
```

Using this array, you can also create matrices and vectors:

$$\begin{bmatrix} \frac{\partial f_1}{\partial P} & \frac{\partial f_1}{\partial G} \\ \frac{\partial f_2}{\partial P} & \frac{\partial f_2}{\partial G} \end{bmatrix}.$$

The command to generate that matrix is:

```
\left[\begin{array}{cc}
\dfrac{\partial f_1}{\partial P} & \dfrac{\partial f_1}{\partial G} \\
\dfrac{\partial f_2}{\partial P} & \dfrac{\partial f_2}{\partial G}
\end{array}\right]
```

Exercise B

1/1 point (graded)

Try to write down which lines would create the following equation:

$$\begin{bmatrix} \frac{dP}{dt} \\ \frac{dG}{dt} \end{bmatrix} = \begin{bmatrix} \left. \frac{\partial f_1}{\partial P} \right|_{(P_0, G_0)} & \left. \frac{\partial f_1}{\partial G} \right|_{(P_0, G_0)} \\ \left. \frac{\partial f_2}{\partial P} \right|_{(P_0, G_0)} & \left. \frac{\partial f_2}{\partial G} \right|_{(P_0, G_0)} \end{bmatrix} \begin{bmatrix} P - P_0 \\ G - G_0 \end{bmatrix}.$$

Did you manage to produce these lines?

☒ Yes; ✓

☐ No.

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You have used 1 of 10 attempts

i Answers are displayed within the problem

As you can see, many special symbols and functions can be typeset by **L^AT_EX**. For a list of symbols, you can use this link: <http://tug.ctan.org/info/symbols/comprehensive/symbols-a4.pdf>. Note that if you want to use special characters, you might have to load an extra package in your preamble

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If you want more information about special things ***L^AT_EX*** can do for your mathematical formulas, check Chapter 3 of this link: <https://tobi.oetiker.ch/lshort/lshort.pdf>.

On the next page, more information on the different uses of formulas in a report.

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English ▼

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